

Supplementary material for the manuscript entitled “Prediction of GO Terms Based on Partitioning PPI Networks into Highly Connected Components”, ID: TCBB-2022-11-0938

Milana Grbić, Branislava Gemović, Savka (ex. Janković) Vračević,
Radoslav Davidović, Dragan Matić, Aleksandar Kartelj

February 19, 2026

This document contains supplementary material supporting the manuscript titled “*Prediction of GO Terms Based on Partitioning PPI Networks into Highly Connected Components*”. The additional content provided herein includes extended experimental results, supplementary tables and figures, detailed examples of the GO term prediction process, as well as further elaboration on the methodology and dataset descriptions.

1 Biological interpretation of metabolite HCD components

As it is stated in main paper, some of the HCD components obtained by partitioning given metabolic networks are analyzed from the biological point of view. More precisely, the process of fatty acid synthesis is recognized in HCD components of all four metabolic networks. This process consists of three consecutive phases: activation, elongation and termination [1, 2]. In each of the phases, some of the intermediates are repeated, so in some organisms the metabolites participating in these phases are denser, whereas in others they are less frequently related. Based on metabolic networks, a network corresponding to one of the phases of fatty acid synthesis is obtained as an HCD component for all four organisms. Figure 1 shows one of the highly related components of the *Saccharomyces cerevisiae* organism. This HCD component represents the termination phase of the fatty acid synthesis process. On the other hand, the HCD component of the metabolic network *Staphylococcus aureus* is a structure which represents the first two phases of the fatty acid synthesis process, i.e., the phases of activation and elongation (Figure 2). Unlike the organism *Staphylococcus aureus*, where the intermediates of the first two phases of the fatty acid synthesis process are connected into one HCD component, the activation phase for the organism *Tuberculosis – Mycobacterium tuberculosis* is separated into a separate component (Figure 3). For the organism *Escherichia coli*, one of the HCD components represents the elongation phase of the fatty acid synthesis process (Figure 4).

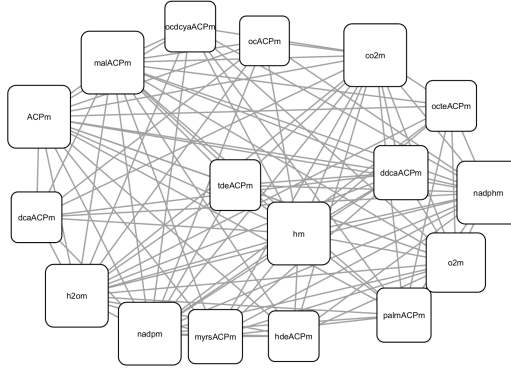


Figure 1: The termination mechanism in fatty acid synthesis in *Saccharomyces cerevisiae* network

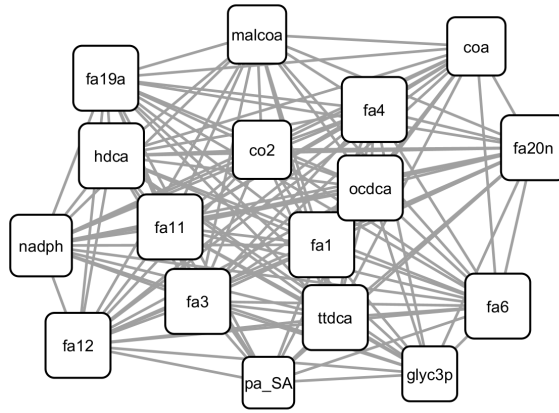


Figure 2: Activation and elongation phases in fatty acid synthesis in *Staphylococcus aureus* network

These findings are in the line with conclusions from paper [3]. More precisely, in paper [3] a network is partitioned into subgraphs, called k -plexes where the degree of each vertex within the k -plex with n vertices is at least $n - k$. In the same study is demonstrated that relaxing the partitioning requirements by varying k can lead to more descriptive and biologically relevant results. Since, the HCD components can be considered as k -plexes with $k = n/2 - 1$, they represent more relaxed partitioning scheme which can lead to more descriptive and biologically relevant results. It is important to mention that in the paper [3] solely to the metabolic network of *Saccharomyces cerevisiae* (yeast) was considered, while in our study, we extend the analysis to include multiple organisms. It is important to note that the k -plex partitioning has not been applied to PPI networks, which prevents us from making a direct comparison with methods of that type.

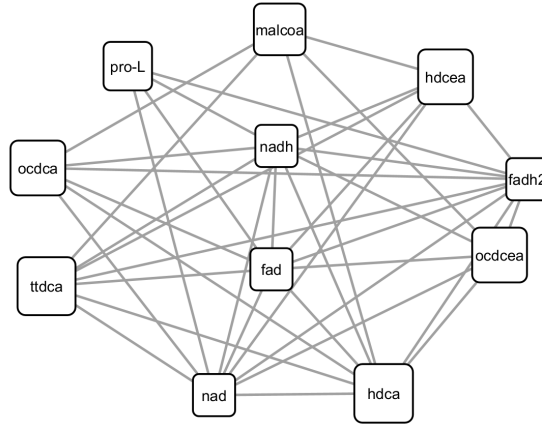


Figure 3: The activation phase in fatty acid synthesis in *Tuberculosis* –*Mycobacterium tuberculosis* network

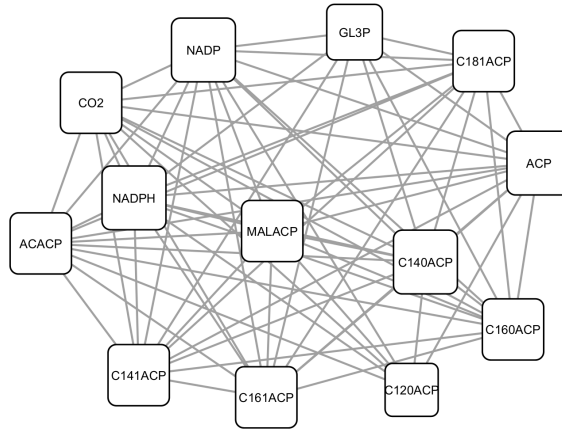


Figure 4: The elongation phase in fatty acid synthesis in *Escherichia coli* network

2 Parameters of VNS and GOTerm prediction methods

In this section, we provide an overview of the key parameters used in the algorithms. A detailed description of each parameter, along with its value is shown in Tables 1 and 2, is provided to ensure clarity and to allow for reproducibility of the experiments.

Table 1: VNS Parameters

Parameter	Description	Value
k_{min}	Minimum size of neighborhood in Shaking	1
k_{max}	Maximum size of neighborhood in Shaking	20
$iter_{total}$	Total iterations performed in VNS	10000
$iter_{without Impr}$	Maximum iterations without improvement before stopping	2000
$prob$	Probability switching to a new solution when objective functions are equal	0.5
max_time	Maximum time allowed for the algorithm to run	4 hours

Table 2: Parameters for GO Prediction Method

Parameter	Value
Number of repetitions	100
Number of different values for Score1	10
Number of different values for Score2	10
Range for Score1	0.001 – 0.05
Step for Score1	0.0054
Range for Score2	0.001 – 0.999
Step for Score2	0.11
Metric used for evaluation	F1 measure

3 Stability and Sensitivity Analysis

Here we provide detailed results of the stability and sensitivity analysis of the VNS-based method across multiple runs.

Stability is evaluated by calculating the **average gap**:

$$\text{AvgGap} = \frac{1}{R} \sum_{i=1}^R \frac{f_i - f_{\text{best}}}{f_{\text{best}}}$$

and the corresponding **standard deviation**:

$$\text{StdGap} = \sqrt{\frac{1}{R} \sum_{i=1}^R \left(\frac{f_i - f_{\text{best}}}{f_{\text{best}}} - \text{AvgGap} \right)^2}$$

across all runs ($R = 10$) for each network instance. Here, f_i denotes the objective value obtained in the i -th run, and f_{best} is the best objective value observed over all runs.

Structural sensibility of the obtained partitions is assessed using the **Adjusted Rand Index (ARI)**. For each pair of runs, ARI measures the similarity between the two node partitions by comparing all pairs of nodes and counting how consistently they are assigned to the same or to different components in both partitions. The index is corrected for chance, so that a value of 1 indicates identical partitions, while a value close to 0 corresponds to agreement expected at random. In our experiments, ARI values are computed for all pairs of runs of the same instance, and the reported value represents the average similarity across runs.

All results are summarized in Table 3.

4 Complexity of VNS algorithm

Complexity analysis is performed only for the local search procedure, since it represents the most computationally expensive part of the overall algorithm.

A single iteration of the local search algorithm may, in the worst case, be executed for each node $i \in V$. For each such node, the algorithm attempts to move i from its current component p ($i \in V_p$) to any other existing component q , as well as to a new singleton component.

Let $\Delta(G)$ denote the maximum degree of a node in G . For each hypothetical move, a partial evaluation of the objective function is performed. This evaluation runs, in the worst case, in time

$$O(\max(|V_p|, |V_q|, \Delta(G))).$$

The term $|V_p|$ arises because the original component must be examined and the correctness values of its nodes updated. The term $|V_q|$ corresponds to updating the correctness values in the destination component. The term $\Delta(G)$ appears because all neighbors of node i must be inspected to determine whether any of them belong to the target component q ; if none do, a new component is formed.

A naive upper bound on the time complexity of one local search iteration is therefore

$$O(|V| \cdot |V| \cdot |V|) = O(|V|^3),$$

which corresponds to the case where there are many components of large cardinality. Although this bound is formally correct as an upper bound, it does not represent a realistic scenario, since it is not possible to simultaneously have many components of large size.

We therefore consider several special cases.

Case 1: A single component. If all nodes belong to a single component, the outer loop iterates over $|V|$ nodes, and for each node only one move is considered, namely moving

Table 3: Stability and sensitivity statistics of the VNS algorithm

instances	AvgGap	StdGap	Mean_ARI	Std_ARI
At-all	0.001449	0.000769	0.8863	0.0171
AT-phys	0.001284	0.00059	0.8822	0.0198
Ce-all	0.001807	0.000783	0.7483	0.0588
Ce-phys	0.000405	0.000265	0.7483	0.0588
Mm-all	0.000947	0.000525	0.8173	0.0263
Mm-phys	0.000372	0.000361	0.8628	0.0202
Sp-all	0.002626	0.001638	0.649	0.0387
Sp-phys	0.00068	0.000632	0.9187	0.023
Ec	0.002275	0.002638	0.9083	0.0518
Sa	0.000584	0.000485	0.9888	0.0032
Tu	0.001954	0.001465	0.8665	0.0465
Sc	0.003324	0.002224	0.8412	0.0447
gs_50	0	0	1	0
gs_80	0	0	0.9824	0.0188
gs_100	0	0	0.99	0.0093
gs_150	0	0	1	0
gs_200	0	0	1	0
gs_500	0	0	1	0
gs_1000	0	0	1	0
gs_2000	0	0	1	0
gs_5000	0	0	1	0
gs_10000	0	0	1	0
gd_50	0.005028	0.001676	0.7453	0.3293
gd_80	0.000251	0.000754	0.9574	0.0852
gd_100	0	0	0.9779	0.0442
gd_150	0	0	1	0
gd_200	0	0	1	0
gd_500	0	0	1	0
gd_1000	0	0	1	0
gd_2000	0	0	1	0
gd_5000	0	0	0.9982	0.0027
gd_10000	0	0	1	0

it to its own singleton component. In this case, the partial evaluation of the objective function runs in time

$$O(\max(|V|, 0, \Delta(G))) = O(|V|),$$

and the total complexity of one iteration of local search is $O(|V|^2)$.

Case 2: All components are singletons. If every node forms its own component, the outer loop runs for $|V|$ iterations, while the inner loop considers $|V| - 1$ possible destination components. The partial evaluation then runs in time $O(\Delta(G))$, yielding an overall complexity of

$$O(|V|^2 \cdot \Delta(G)).$$

Intermediate cases. All intermediate cases are cheaper in terms of time complexity than $O(|V|^2 \cdot \Delta(G))$. For example, suppose there are m components of approximately equal size $|V|/m$, where m is a constant smaller than $|V|$ and larger than 1. The outer loop iterates over $|V|$ nodes, the inner loop over m components, and the partial objective evaluation runs in time

$$O\left(\max\left(\frac{|V|}{m}, \Delta(G)\right)\right).$$

If $\Delta(G) \geq |V|/m$, the total complexity is $O(|V| \cdot m \cdot \Delta(G)) \leq O(|V|^2 \cdot \Delta(G))$. Otherwise, if $|V|/m > \Delta(G)$, the complexity becomes

$$O\left(|V| \cdot m \cdot \frac{|V|}{m}\right) = O(|V|^2),$$

which is again upper-bounded by $O(|V|^2 \cdot \Delta(G))$.

We therefore conclude that the time complexity of a single iteration of the local search algorithm is

$$O(|V|^2 \cdot \Delta(G)).$$

The total number of local search iterations cannot be determined in advance, since the procedure is executed repeatedly as long as improvements are found. Therefore, we estimate the number of local search iterations empirically by performing a single run on artificial random instances with up to 1000 nodes. The obtained results are reported in Table 4.

The first four columns have been explained previously. Regarding the remaining columns, *VNS iter.* denotes the number of iterations of the main VNS loop, which is unknown in advance but is upper bounded by a user-defined control parameter; *LS iter.* represents the total number of local search iterations, which is also unknown beforehand; *Iter. ratio* gives the ratio between these two values; *Iter. cost* corresponds to an upper bound on the cost of a single local search iteration; and *Cost* provides an estimate of the total computational cost of all local search iterations, computed as the product of *LS iter.* and *Iter. cost*. The last column reports the actual execution time.

As shown below the table, the estimated total cost is highly correlated with the realized execution time ($\rho \approx 0.98$), which confirms the validity of the proposed upper bound on the complexity of a single local search iteration. Concerning *LS iter.*, as already noted, it cannot be directly bounded in advance, since it depends on the specific problem instance and its internal structure. However, inspection of the *Iter. ratio* column reveals that

higher graph density leads to larger ratios. In particular, for sparse instances, the number of local search iterations is up to approximately 1.3 times the number of VNS iterations, whereas for dense instances this factor increases to about 4.5.

Nevertheless, the total number of local search iterations exhibits an approximately linear dependence on the maximum number of VNS iterations, which is a user-specified control parameter. Consequently, the overall time complexity can be expressed as the product of the maximum number of VNS iterations and $O(|V|^2 \cdot \Delta(G))$.

Table 4: Empirical analysis of estimated and realized complexity for artificial instances with up to 1000 nodes.

Instance	n	m	$density$	$\Delta(G)$	VNS iter.	LS iter.	Iter. ratio	Iter. cost	Cost	$t[s]$
gs_50	50	102	0.083	9	50000	52699	1.05	22500	1.19×10^9	6.43
gs_80	80	149	0.047	9	50011	61156	1.22	57600	3.52×10^9	7.65
gs_100	100	312	0.063	15	50127	64350	1.28	150000	9.65×10^9	11.67
gs_150	150	0.078	875	21	50003	57202	1.14	472500	2.70×10^{10}	19.93
gs_200	200	0.071	1410	23	50006	50006	1.00	920000	4.60×10^{10}	25.33
gs_500	500	4035	0.032	26	50003	50003	1.00	6500000	3.25×10^{11}	103.84
gs_1000	1000	8409	0.017	27	50001	50002	1.00	27000000	1.35×10^{12}	343.88
gd_50	50	244	0.199	16	57696	150595	2.61	40000	6.02×10^9	14.32
gd_80	80	527	0.167	21	54946	219831	4.00	134400	2.95×10^{10}	32.91
gd_100	100	858	0.173	28	55540	250034	4.50	280000	7.00×10^{10}	60.09
gd_150	150	2149	0.192	47	50070	100477	2.01	1057500	1.06×10^{11}	54.34
gd_200	200	3818	0.192	59	50247	90951	1.81	2360000	2.15×10^{11}	72.67
gd_500	500	18381	0.147	100	50599	85956	1.70	25000000	2.15×10^{12}	298.79
gd_1000	1000	63406	0.127	163	50469	102441	2.03	163000000	1.67×10^{13}	1219.77
					$\rho(density, Iter.ratio) \approx 0.69$			$\rho(Cost, t[s]) \approx 0.98$		

5 Addtional evalutaion of HCDAM method

In this section, we provide a detailed overview of the external temporal validation results for the H. sapiens dataset. The tables 5, 6 and 7 list the Recall, Precision, and F1 scores obtained for different combinations of DINGO and STRING cutoff parameters across the three GO subontologies, BPO, MFO, and CCO, respectively. These results complement the summary presented in the main text and allow for a more comprehensive assessment of the performance of HCDAM under varying parameter settings.

Each table presents the following columns, in order: DINGO cutoff, STRING cutoff, True Positives (TP), False Positives (FP), False Negatives (FN), Recall, Precision, and F1 score. In each table, the cutoff parameter combination that was reported as optimal in the main text is highlighted in bold.

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
1.00E-05	0.222777778	20280	156160	111676	0.153688	0.11494	0.131519
1.00E-05	0.333666667	16740	109181	115216	0.12686	0.13294	0.129829
1.00E-05	0.111888889	23215	209900	108741	0.17593	0.099586	0.127181
1.00E-05	0.444555556	13620	75455	118336	0.103216	0.152905	0.123241
1.00E-05	0.001	23594	231112	108362	0.178802	0.092632	0.122039
0.005564444	0.555444444	17474	140089	114482	0.132423	0.110902	0.120711
0.005564444	0.444555556	20583	191567	111373	0.155984	0.097021	0.119632
0.011118889	0.555444444	18938	168961	113018	0.143518	0.100788	0.118416
0.005564444	0.333666667	24155	259011	107801	0.183053	0.085303	0.116375
0.011118889	0.444555556	22139	229161	109817	0.167776	0.088098	0.115531
0.016673333	0.555444444	20184	199253	111772	0.15296	0.091981	0.11488
0.011118889	0.666333333	15535	123649	116421	0.117729	0.111615	0.11459
0.005564444	0.666333333	14153	102044	117803	0.107255	0.121802	0.114067
1.00E-05	0.555444444	11189	53077	120767	0.084793	0.174105	0.114044
0.016673333	0.666333333	16777	148322	115179	0.127141	0.101618	0.112956
0.022227778	0.555444444	21287	226425	110669	0.161319	0.085934	0.112135
0.016673333	0.444555556	23580	266463	108376	0.178696	0.081298	0.111754
0.011118889	0.333666667	25829	306503	106127	0.195739	0.07772	0.111263
0.027782222	0.555444444	22115	243508	109841	0.167594	0.083257	0.111248
0.022227778	0.666333333	17793	172092	114163	0.13484	0.093704	0.11057
0.027782222	0.666333333	18565	185367	113391	0.140691	0.091035	0.110543
0.022227778	0.444555556	24843	298862	107113	0.188267	0.076746	0.109042
0.005564444	0.222777778	28388	362130	103568	0.215132	0.072693	0.108668
0.033336667	0.555444444	23160	271501	108796	0.175513	0.078599	0.108575
0.027782222	0.444555556	25851	320616	106105	0.195906	0.074613	0.108068
0.033336667	0.666333333	19533	210330	112423	0.148027	0.084977	0.107971
0.016673333	0.777222222	14229	117432	117727	0.107831	0.108073	0.107952
0.016673333	0.333666667	27673	355485	104283	0.209714	0.072223	0.107444
0.011118889	0.777222222	12916	95750	119040	0.097881	0.11886	0.107355
0.027782222	0.777222222	15979	150885	115977	0.121093	0.095761	0.106947
0.038891111	0.555444444	23928	292317	108028	0.181333	0.075663	0.106774
0.038891111	0.666333333	20269	227605	111687	0.153604	0.081771	0.106727
0.022227778	0.777222222	15214	139430	116742	0.115296	0.098381	0.106169
0.044445556	0.555444444	24609	311688	107347	0.186494	0.073176	0.10511
0.033336667	0.444555556	26878	353112	105078	0.203689	0.070733	0.105003
0.044445556	0.666333333	20806	244342	111150	0.157674	0.078469	0.104789
0.033336667	0.777222222	16881	173574	115075	0.127929	0.088635	0.104717
0.022227778	0.333666667	29017	395316	102939	0.219899	0.068383	0.104323
0.038891111	0.777222222	17568	188827	114388	0.133135	0.085118	0.103845
0.05	0.555444444	25111	328469	106845	0.190298	0.071019	0.103436
0.011118889	0.222777778	30434	426241	101522	0.230637	0.066643	0.103406
0.05	0.666333333	21291	258968	110665	0.161349	0.075969	0.1033

0.027782222	0.333666667	30164	423867	101792	0.228591	0.066436	0.102951
0.005564444	0.777222222	11448	79052	120508	0.086756	0.126497	0.102924
0.038891111	0.444555556	27654	378365	104302	0.20957	0.06811	0.102808
0.044445556	0.777222222	18113	204161	113843	0.137265	0.08149	0.102267
0.05	0.777222222	18627	217332	113329	0.141161	0.078942	0.101257
0.044445556	0.444555556	28417	401652	103539	0.215352	0.066075	0.101124
0.027782222	0.888111111	13487	121969	118469	0.102208	0.099567	0.100871
0.033336667	0.333666667	31384	463650	100572	0.237837	0.063398	0.10011
0.033336667	0.888111111	14391	141365	117565	0.109059	0.092395	0.100038
0.038891111	0.888111111	15026	154236	116930	0.113871	0.088774	0.099768
0.016673333	0.222777778	32731	492121	99225	0.248045	0.062362	0.099667
0.05	0.444555556	29057	422186	102899	0.220202	0.064393	0.099647
0.022227778	0.888111111	12775	112214	119181	0.096813	0.102209	0.099438
0.005564444	0.111888889	32143	483448	99813	0.243589	0.062342	0.099276
1.00E-05	0.666333333	8837	38646	123119	0.066969	0.186109	0.098496
0.044445556	0.888111111	15535	168238	116421	0.117729	0.084534	0.098407
0.016673333	0.888111111	11659	93406	120297	0.088355	0.110969	0.098379
0.038891111	0.333666667	32412	496476	99544	0.245627	0.061283	0.098093
0.05	0.888111111	16008	179006	115948	0.121313	0.082086	0.097917
0.022227778	0.222777778	34400	547170	97556	0.260693	0.05915	0.096423
0.044445556	0.333666667	33231	525886	98725	0.251834	0.059435	0.096172
0.011118889	0.888111111	10477	76547	121479	0.079398	0.120392	0.095689
0.027782222	0.222777778	35868	589303	96088	0.271818	0.057373	0.094748
0.011118889	0.111888889	34830	571279	97126	0.263952	0.057465	0.094382
0.05	0.333666667	33847	552185	98109	0.256502	0.057756	0.094283
0.005564444	0.001	32655	534766	99301	0.247469	0.05755	0.093383
0.033336667	0.222777778	37317	642553	94639	0.282799	0.054888	0.091933
0.005564444	0.888111111	9196	63110	122760	0.06969	0.127182	0.090041
0.016673333	0.111888889	37366	662228	94590	0.28317	0.053411	0.089871
0.038891111	0.222777778	38458	686586	93498	0.291446	0.053042	0.08975
0.011118889	0.001	35299	629710	96657	0.267506	0.05308	0.088584
0.044445556	0.222777778	39375	727668	92581	0.298395	0.051333	0.087597
0.022227778	0.111888889	39323	735021	92633	0.298001	0.050782	0.086777
0.05	0.222777778	40095	765481	91861	0.303851	0.049772	0.085533
0.027782222	0.111888889	41171	798814	90785	0.312006	0.049014	0.084719
0.016673333	0.001	37823	726750	94133	0.286633	0.049469	0.084377
0.033336667	0.111888889	42793	868291	89163	0.324297	0.046969	0.082054
0.022227778	0.001	39762	801994	92194	0.301328	0.047237	0.081671
1.00E-05	0.777222222	6914	30914	125042	0.052396	0.182775	0.081445
0.038891111	0.111888889	44298	927674	87658	0.335703	0.045575	0.080255
0.027782222	0.001	41617	870302	90339	0.315385	0.045637	0.079736
0.044445556	0.111888889	45392	983046	86564	0.343993	0.044137	0.078235
0.033336667	0.001	43238	941672	88718	0.32767	0.0439	0.077427
0.05	0.111888889	46339	1035462	85617	0.35117	0.042835	0.076356

0.038891111	0.001	44726	1003382	87230	0.338946	0.042673	0.075803
0.044445556	0.001	45796	1060986	86160	0.347055	0.041378	0.07394
0.05	0.001	46721	1116852	85235	0.354065	0.040153	0.072127
1.00E-05	0.888111111	5038	24066	126918	0.038179	0.173103	0.062561
0.05	0.999	2153	11485	129803	0.016316	0.157868	0.029575
0.044445556	0.999	2058	10245	129898	0.015596	0.167276	0.028532
0.038891111	0.999	1842	7807	130114	0.013959	0.190901	0.026016
0.033336667	0.999	1705	6475	130251	0.012921	0.208435	0.024334
0.027782222	0.999	1425	4707	130531	0.010799	0.232387	0.020639
0.022227778	0.999	1219	3721	130737	0.009238	0.246761	0.017809
0.016673333	0.999	975	2636	130981	0.007389	0.270008	0.014384
0.011118889	0.999	756	1617	131200	0.005729	0.318584	0.011256
0.005564444	0.999	510	912	131446	0.003865	0.35865	0.007647
1.00E-05	0.999	82	38	131874	6.21E-04	0.683333	0.001242

Table 5: Detailed external validation results for *H. sapiens* – Biological Process Ontology

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
1.00E-05	0.222777778	4381	23766	26549	0.141642	0.155647	0.148315
0.005564444	0.333666667	5028	34347	25902	0.162561	0.127695	0.143034
1.00E-05	0.111888889	5268	37892	25662	0.17032	0.122057	0.142205
0.011118889	0.333666667	5484	40899	25446	0.177304	0.118233	0.141865
0.005564444	0.222777778	6260	51381	24670	0.202392	0.108603	0.141356
0.011118889	0.444555556	4454	28182	26476	0.144003	0.136475	0.140138
0.016673333	0.333666667	5785	46660	25145	0.187035	0.110306	0.138771
0.016673333	0.444555556	4704	32566	26226	0.152085	0.126214	0.137947
0.011118889	0.222777778	6752	61419	24178	0.218299	0.099045	0.136265
0.022227778	0.444555556	4851	35624	26079	0.156838	0.119852	0.135873
0.022227778	0.333666667	5939	50744	24991	0.192014	0.104776	0.135573
0.027782222	0.444555556	5062	38702	25868	0.16366	0.115666	0.13554
0.005564444	0.444555556	3928	23187	27002	0.126996	0.144864	0.135343
0.033336667	0.444555556	5163	40259	25767	0.166925	0.113667	0.135242
1.00E-05	0.333666667	3316	14900	27614	0.10721	0.182038	0.134945
1.00E-05	0.001	5450	44625	25480	0.176204	0.108837	0.13456
0.027782222	0.333666667	6138	54692	24792	0.198448	0.100904	0.133784
0.033336667	0.333666667	6261	56850	24669	0.202425	0.099206	0.133155
0.038891111	0.444555556	5251	43228	25679	0.16977	0.108315	0.132252
0.016673333	0.222777778	7074	69757	23856	0.22871	0.092072	0.131291
0.044445556	0.444555556	5311	45194	25619	0.17171	0.105158	0.130435
0.05	0.444555556	5413	47160	25517	0.175008	0.102962	0.129648
0.038891111	0.333666667	6343	60730	24587	0.205076	0.094569	0.129445
0.022227778	0.222777778	7262	75411	23668	0.234788	0.08784	0.127849
0.033336667	0.555444444	4116	29343	26814	0.133075	0.123016	0.127848
0.044445556	0.333666667	6419	63444	24511	0.207533	0.09188	0.12737
0.027782222	0.555444444	4015	28149	26915	0.129809	0.124829	0.12727
0.011118889	0.555444444	3436	19684	27494	0.11109	0.148616	0.127142
0.016673333	0.555444444	3669	23271	27261	0.118623	0.136192	0.126801
0.022227778	0.555444444	3823	25649	27107	0.123602	0.129716	0.126585
0.038891111	0.555444444	4209	31653	26721	0.136081	0.117367	0.126033
0.05	0.333666667	6511	66003	24419	0.210508	0.08979	0.125885
0.005564444	0.111888889	7172	75933	23758	0.231878	0.0863	0.125786
0.027782222	0.222777778	7489	80870	23441	0.242127	0.084757	0.125561
0.05	0.555444444	4386	34578	26544	0.141804	0.112565	0.125504
0.044445556	0.555444444	4279	33132	26651	0.138345	0.114378	0.125225
0.033336667	0.222777778	7624	83739	23306	0.246492	0.083447	0.124684
0.038891111	0.222777778	7721	88854	23209	0.249628	0.079948	0.121109
0.011118889	0.111888889	7749	90602	23181	0.250533	0.078789	0.119878
0.044445556	0.222777778	7822	92686	23108	0.252894	0.077825	0.119022
0.005564444	0.555444444	2965	15968	27965	0.095862	0.156605	0.118926
0.005564444	0.001	7331	86383	23599	0.237019	0.078227	0.117631

0.05	0.222777778	7924	96382	23006	0.256191	0.075969	0.117188
0.05	0.666333333	3494	26371	27436	0.112965	0.116993	0.114944
0.016673333	0.111888889	8143	102856	22787	0.263272	0.073361	0.114748
0.033336667	0.666333333	3228	22201	27702	0.104365	0.126942	0.114551
0.038891111	0.666333333	3323	24077	27607	0.107436	0.121277	0.113938
0.044445556	0.666333333	3393	25253	27537	0.109699	0.118446	0.113905
0.027782222	0.666333333	3135	21213	27795	0.101358	0.128758	0.113427
1.00E-05	0.444555556	2428	9526	28502	0.0785	0.203112	0.113236
0.011118889	0.001	7909	101440	23021	0.255706	0.072328	0.112761
0.022227778	0.111888889	8429	111472	22501	0.272519	0.0703	0.111767
0.022227778	0.666333333	2953	19139	27977	0.095474	0.133668	0.111388
0.016673333	0.666333333	2817	17209	28113	0.091077	0.140667	0.110566
0.027782222	0.111888889	8709	118849	22221	0.281571	0.068275	0.109901
0.033336667	0.111888889	8899	122788	22031	0.287714	0.067577	0.109447
0.011118889	0.666333333	2610	14295	28320	0.084384	0.154392	0.109125
0.016673333	0.001	8296	114657	22634	0.268219	0.067473	0.107822
0.038891111	0.111888889	9077	130421	21853	0.293469	0.065069	0.10652
0.022227778	0.001	8574	123528	22356	0.277207	0.064904	0.105182
0.05	0.777222222	2892	21201	28038	0.093501	0.120035	0.10512
0.044445556	0.111888889	9220	135964	21710	0.298092	0.063506	0.104705
0.033336667	0.777222222	2669	17779	28261	0.086292	0.130526	0.103897
0.044445556	0.777222222	2805	20268	28125	0.090689	0.121571	0.103883
0.027782222	0.001	8851	131078	22079	0.286162	0.063254	0.103606
0.038891111	0.777222222	2742	19353	28188	0.088652	0.1241	0.103423
0.05	0.111888889	9364	140931	21566	0.302748	0.062304	0.103341
0.033336667	0.001	9043	135103	21887	0.29237	0.062735	0.103304
0.005564444	0.666333333	2248	11320	28682	0.07268	0.165684	0.101038
0.038891111	0.001	9214	142792	21716	0.297898	0.060616	0.100735
0.027782222	0.777222222	2538	16923	28392	0.082056	0.130415	0.100732
0.022227778	0.777222222	2414	15178	28516	0.078047	0.137221	0.099501
0.044445556	0.001	9360	148394	21570	0.302619	0.059333	0.099213
0.05	0.001	9501	153423	21429	0.307177	0.058316	0.098022
0.016673333	0.777222222	2284	13556	28646	0.073844	0.144192	0.097669
0.05	0.888111111	2441	16945	28489	0.07892	0.125916	0.097027
0.044445556	0.888111111	2371	16178	28559	0.076657	0.127824	0.095839
0.033336667	0.888111111	2262	14174	28668	0.073133	0.137625	0.095512
0.038891111	0.888111111	2321	15421	28609	0.07504	0.13082	0.095373
0.011118889	0.777222222	2047	11178	28883	0.066182	0.154783	0.092719
1.00E-05	0.555444444	1794	5995	29136	0.058002	0.230325	0.092668
0.027782222	0.888111111	2128	13466	28802	0.068801	0.136463	0.09148
0.022227778	0.888111111	2019	11918	28911	0.065276	0.144866	0.089999
0.016673333	0.888111111	1900	10609	29030	0.061429	0.151891	0.087479
0.005564444	0.777222222	1747	8587	29183	0.056482	0.169054	0.084674
0.011118889	0.888111111	1710	8611	29220	0.055286	0.165682	0.082907

1.00E-05	0.666333333	1382	4133	29548	0.044682	0.250589	0.07584
0.005564444	0.888111111	1452	6569	29478	0.046945	0.181025	0.074555
1.00E-05	0.777222222	1049	3054	29881	0.033915	0.255667	0.059886
1.00E-05	0.888111111	827	2280	30103	0.026738	0.266173	0.048594
0.05	0.999	220	927	30710	0.007113	0.191805	0.013717
0.044445556	0.999	202	854	30728	0.006531	0.191288	0.012631
0.038891111	0.999	199	805	30731	0.006434	0.198207	0.012463
0.033336667	0.999	178	730	30752	0.005755	0.196035	0.011182
0.027782222	0.999	167	659	30763	0.005399	0.202179	0.010518
0.022227778	0.999	118	418	30812	0.003815	0.220149	0.0075
0.016673333	0.999	82	323	30848	0.002651	0.202469	0.005234
0.011118889	0.999	45	255	30885	0.001455	0.15	0.002882
0.005564444	0.999	38	214	30892	0.001229	0.150794	0.002437
1.00E-05	0.999	2	4	30928	6.47E-05	0.333333	1.29E-04

Table 6: Detailed external validation results for H. sapiens – Molecular Function Ontology

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
0.016673	0.111889	15032	66904	40824	0.269121	0.18346	0.218184
1.00E-05	0.001	10116	27763	45740	0.181109	0.267061	0.215843
0.011119	0.111889	14109	60771	41747	0.252596	0.188421	0.21584
0.016673	0.222778	12910	50888	42946	0.23113	0.202357	0.215789
0.022228	0.111889	15636	73455	40220	0.279934	0.175506	0.215748
0.038891	0.222778	14677	65907	41179	0.262765	0.182133	0.215142
0.022228	0.222778	13476	56123	42380	0.241263	0.193623	0.214834
0.011119	0.222778	12298	46439	43558	0.220173	0.209374	0.214638
0.033337	0.222778	14194	62264	41662	0.254118	0.185644	0.21455
0.027782	0.111889	16100	78132	39756	0.288241	0.170855	0.214541
0.027782	0.222778	13866	59630	41990	0.248245	0.188663	0.214392
1.00E-05	0.111889	9760	25475	46096	0.174735	0.276997	0.214291
0.033337	0.111889	16481	81655	39375	0.295062	0.16794	0.21405
0.044446	0.222778	14930	68764	40926	0.267294	0.178388	0.213973
0.05	0.222778	15212	71379	40644	0.272343	0.175676	0.213581
0.038891	0.111889	16990	86408	38866	0.304175	0.164317	0.21337
0.005564	0.111889	12935	52594	42921	0.231578	0.197394	0.213124
0.016673	0.001	15318	73214	40538	0.274241	0.173022	0.212178
0.044446	0.111889	17343	90340	38513	0.310495	0.161056	0.212096
0.038891	0.333667	12449	49238	43407	0.222877	0.201809	0.21182
0.033337	0.333667	12132	46657	43724	0.217201	0.206365	0.211645
0.005564	0.222778	11362	40205	44494	0.203416	0.220335	0.211538
0.027782	0.333667	11882	44748	43974	0.212726	0.209818	0.211262
0.05	0.111889	17660	93907	38196	0.31617	0.158291	0.210963
0.05	0.333667	12872	53380	42984	0.23045	0.194288	0.21083
0.044446	0.333667	12623	51323	43233	0.225992	0.197401	0.210731
0.016673	0.333667	11100	38425	44756	0.198725	0.224129	0.210664
0.022228	0.333667	11555	42336	44301	0.206871	0.214414	0.210575
0.011119	0.001	14405	66787	41451	0.257895	0.177419	0.210218
0.022228	0.001	15907	80055	39949	0.284786	0.165764	0.209554
0.027782	0.001	16382	85019	39474	0.29329	0.161557	0.208347
0.005564	0.001	13224	58067	42632	0.236752	0.185493	0.208011
0.033337	0.001	16774	88695	39082	0.300308	0.159042	0.207953
0.011119	0.333667	10540	34990	45316	0.1887	0.231496	0.207918
0.038891	0.001	17261	93537	38595	0.309027	0.155788	0.207148
0.044446	0.001	17603	97528	38253	0.31515	0.152895	0.205899
0.038891	0.444556	10736	38080	45120	0.192209	0.219928	0.205136
0.05	0.001	17919	101174	37937	0.320807	0.150462	0.204848
0.033337	0.444556	10478	36155	45378	0.18759	0.224691	0.204471
0.005564	0.333667	9826	30575	46030	0.175917	0.243212	0.204162
0.044446	0.444556	10839	39637	45017	0.194053	0.214736	0.203871
0.027782	0.444556	10261	34584	45595	0.183705	0.22881	0.203791

0.05	0.444556	11009	41210	44847	0.197096	0.210824	0.203729
0.022228	0.444556	9933	32591	45923	0.177832	0.233586	0.201931
1.00E-05	0.222778	8514	20107	47342	0.152428	0.297474	0.20157
0.016673	0.444556	9544	29507	46312	0.170868	0.244398	0.201123
0.011119	0.444556	9017	26736	46839	0.161433	0.252203	0.196858
0.038891	0.555444	9252	29673	46604	0.16564	0.237688	0.195229
0.05	0.555444	9512	32183	46344	0.170295	0.228133	0.195016
0.044446	0.555444	9377	30946	46479	0.167878	0.232547	0.194991
0.033337	0.555444	9000	27965	46856	0.161129	0.243474	0.193922
0.027782	0.555444	8839	26631	47017	0.158246	0.249197	0.19357
0.022228	0.555444	8582	25049	47274	0.153645	0.255181	0.191804
0.005564	0.444556	8351	23223	47505	0.149509	0.26449	0.191033
0.016673	0.555444	8195	22599	47661	0.146717	0.266123	0.189152
0.05	0.666333	8400	25982	47456	0.150387	0.244314	0.186174
0.038891	0.666333	8127	23842	47729	0.145499	0.254215	0.185073
0.044446	0.666333	8229	24912	47627	0.147325	0.248303	0.184928
0.011119	0.555444	7773	20493	48083	0.139161	0.274995	0.184803
0.033337	0.666333	7906	22398	47950	0.141543	0.26089	0.183519
1.00E-05	0.333667	7211	15739	48645	0.1291	0.314205	0.183006
0.027782	0.666333	7734	21202	48122	0.138463	0.26728	0.182423
0.022228	0.666333	7525	19936	48331	0.134721	0.274025	0.180635
0.005564	0.555444	7216	17994	48640	0.129189	0.286236	0.178028
0.016673	0.666333	7192	17876	48664	0.12876	0.2869	0.177747
0.011119	0.666333	6881	16445	48975	0.123192	0.294993	0.173802
0.05	0.777222	7243	20844	48613	0.129673	0.257877	0.172569
0.038891	0.777222	7008	18943	48848	0.125465	0.270047	0.17133
0.044446	0.777222	7072	19926	48784	0.126611	0.261945	0.17071
0.033337	0.777222	6808	17609	49048	0.121885	0.278822	0.169621
0.027782	0.777222	6664	16633	49192	0.119307	0.286045	0.168383
0.022228	0.777222	6498	15609	49358	0.116335	0.293934	0.166694
0.005564	0.666333	6374	14506	49482	0.114115	0.305268	0.166128
1.00E-05	0.444556	6105	12125	49751	0.109299	0.334888	0.164808
0.016673	0.777222	6165	14134	49691	0.110373	0.30371	0.161907
0.011119	0.777222	5857	12893	49999	0.104859	0.312373	0.157012
0.05	0.888111	6213	17100	49643	0.111232	0.266504	0.156955
0.038891	0.888111	6019	15378	49837	0.107759	0.281301	0.155826
0.044446	0.888111	6080	16311	49776	0.108851	0.271538	0.155405
0.033337	0.888111	5848	14123	50008	0.104698	0.292825	0.154246
0.027782	0.888111	5696	13243	50160	0.101977	0.300755	0.15231
1.00E-05	0.555444	5402	9874	50454	0.096713	0.353627	0.151887
0.022228	0.888111	5548	12389	50308	0.099327	0.309305	0.150367
0.005564	0.777222	5396	11285	50460	0.096606	0.323482	0.148779
0.016673	0.888111	5194	11175	50662	0.092989	0.317307	0.143828
0.011119	0.888111	4969	10194	50887	0.088961	0.327706	0.139934

1.00E-05	0.666333	4787	8065	51069	0.085703	0.372471	0.139343
0.005564	0.888111	4529	8819	51327	0.081084	0.339302	0.130888
1.00E-05	0.777222	3956	6216	51900	0.070825	0.388911	0.119828
1.00E-05	0.888111	3239	4615	52617	0.057988	0.412401	0.101679
0.05	0.999	1360	1455	54496	0.024348	0.483126	0.04636
0.044446	0.999	1322	1364	54534	0.023668	0.492182	0.045164
0.038891	0.999	1283	1295	54573	0.02297	0.497673	0.043913
0.033337	0.999	1214	1201	54642	0.021734	0.502692	0.041667
0.027782	0.999	1106	1094	54750	0.019801	0.502727	0.038101
0.022228	0.999	1033	967	54823	0.018494	0.5165	0.035709
0.016673	0.999	680	732	55176	0.012174	0.481586	0.023748
0.011119	0.999	583	584	55273	0.010438	0.499572	0.020448
0.005564	0.999	394	336	55462	0.007054	0.539726	0.013926
1.00E-05	0.999	7	11	55849	1.25E-04	0.388889	2.51E-04

Table 7: Detailed external validation results for H. sapiens – Cellular Component Ontology

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